Trophy Tracker - Cross-Platform Video Game Achievement Log

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Project URL: http://classwork.engr.oregonstate.edu:5206

Overview

With the wide range of gaming platforms, it is difficult for players to view their trophies and achievements in a centralized place. Trophy Tracker is a cross-platform video game achievement tracking system that manages trophy and achievement data across multiple gaming platforms, including Steam, PlayStation, Xbox, and Nintendo Switch. Its purpose is to give users who play games on several different platforms a centralized location where they can view all of the trophies they have unlocked. The database tracks 500+ games with an average of 50 achievements per game, totaling over 25,000 individual achievements. The system serves 1,000+ gamers who collectively unlock approximately 5,000 achievements per week. Each game is available on an average of 2-3 platforms, and players maintain collections averaging 30-40 games.

Database Outline

Games: Records details about video games tracked in the system.

- gameID: int, AI, UQ, NN, PK
- title: varchar(255), NN
- developer: varchar(255), NN
- releaseYear: int, NN

Relationship(s):

- 1:M relationship with Achievements, implemented with gameID as FK
- M:M relationship with Platforms through gamePlatforms
- M:M relationship with Players through playerGames

Players: Records details about users tracking their gaming achievements.

- playerID: int, AI, UQ, NN, PK
- username: varchar(100), UQ, NN
- email: varchar(255), UQ, NN

Relationship(s):

- M:M relationship with Achievements through playerAchievements
- M:M relationship with Games through playerGames

Achievements: Records individual achievements/trophies for each game on each platform. Since games can have different achievement lists on different platforms, achievements are tied

to both a specific game and a specific platform. This will also let players who own the same game on several different platforms keep track of the achievements they've unlocked for each specific copy of that game.

- achievementID: int, AI, UQ, NN, PK
- gameID: int, NN, FK
- platformID: int, NN, FK
- achievementName: varchar(255), NN
- description: text, NN
- isHidden: boolean, default false -- some achievement details are hidden until unlocked
- rarityPercentage: decimal(5,2), NULL -- percentage of players who unlocked this, sourced from the platform to avoid calculation overhead.

Relationship(s):

- M:1 relationship with Games
- M:M relationship with Players through playerAchievements

Platforms: Records the different gaming platforms tracked in the system

- platformID: int, AI, UQ, NN, PK
- name: varchar(100), UQ, NN -- e.g. Steam, PlayStation 5, Xbox Series X, Nintendo Switch
- manufacturer: varchar(100), not NULL -- e.g., Valve, Sony, Microsoft, Nintendo

Relationship(s):

- M:M relationship with Games through gamePlatforms

playerGames: Records which player owns which game on which platform. Implements the M:M relationship between Players and Games.

- playerGameID: Int, AI, UQ, NN, PK
- gameID: int, NN, FK
- playerID: int, NN, FK

Relationship(s):

- 1:M relationship with Players
- 1:M relationship with Games

playerAchievements: Records which player unlocked which achievement. Implements the M:M relationship between Players and Achievements.

- playerAchievementID: int, AI, UQ, NN, PK
- playerID: int, NN, FK
- achievementID: int, NN, FK
- dateAchieved: date, NN

Relationship(s):

- 1:M relationship with Players
- 1:M relationship with Achievements

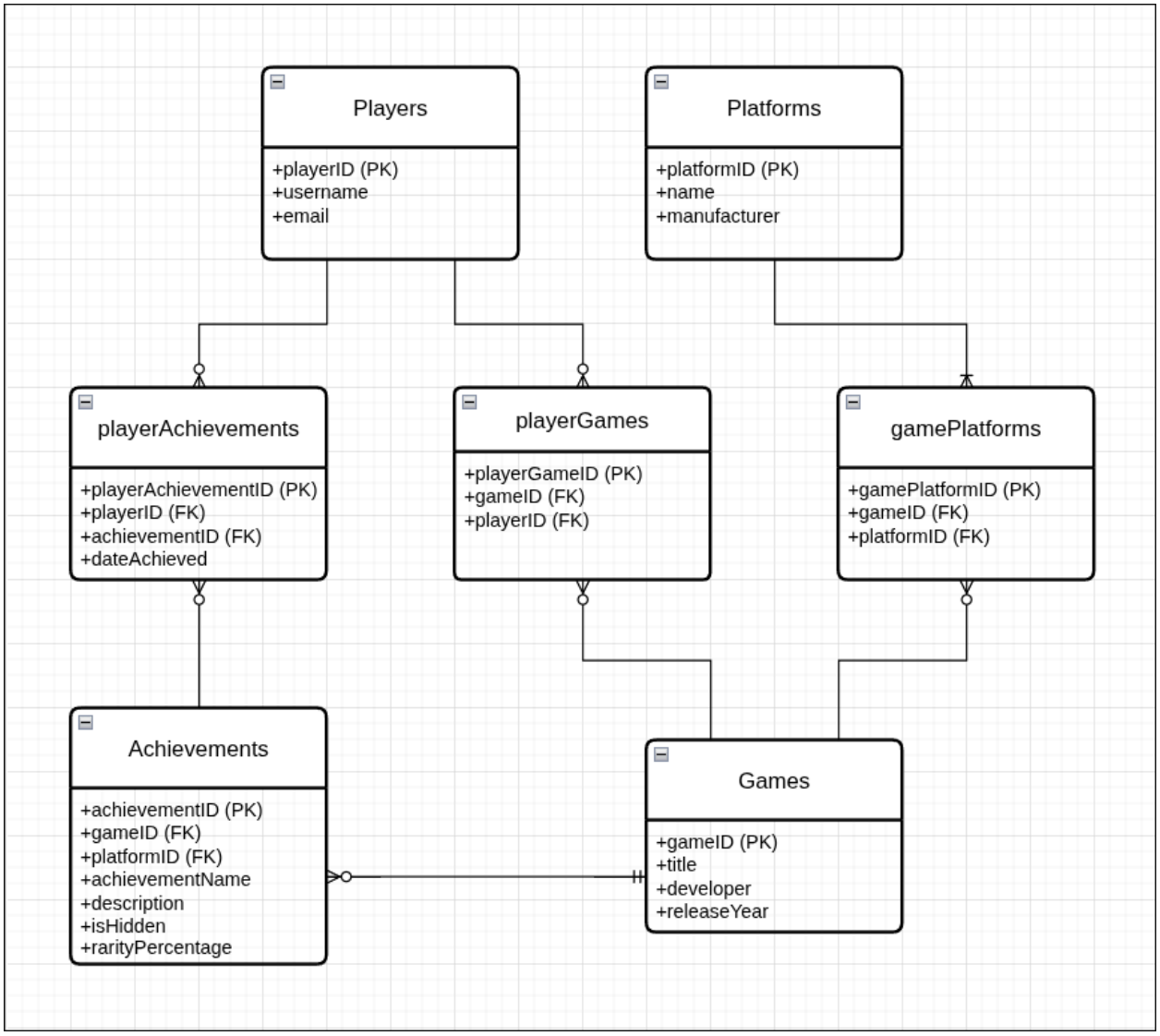
gamePlatforms: Records where a game is available on a platform. Implements the M:M relationship between Games and Platforms.

- gamePlatformID: Int, AI, UQ, NN, PK
- gameID: int, NN, FK
- platformID: int, NN, FK

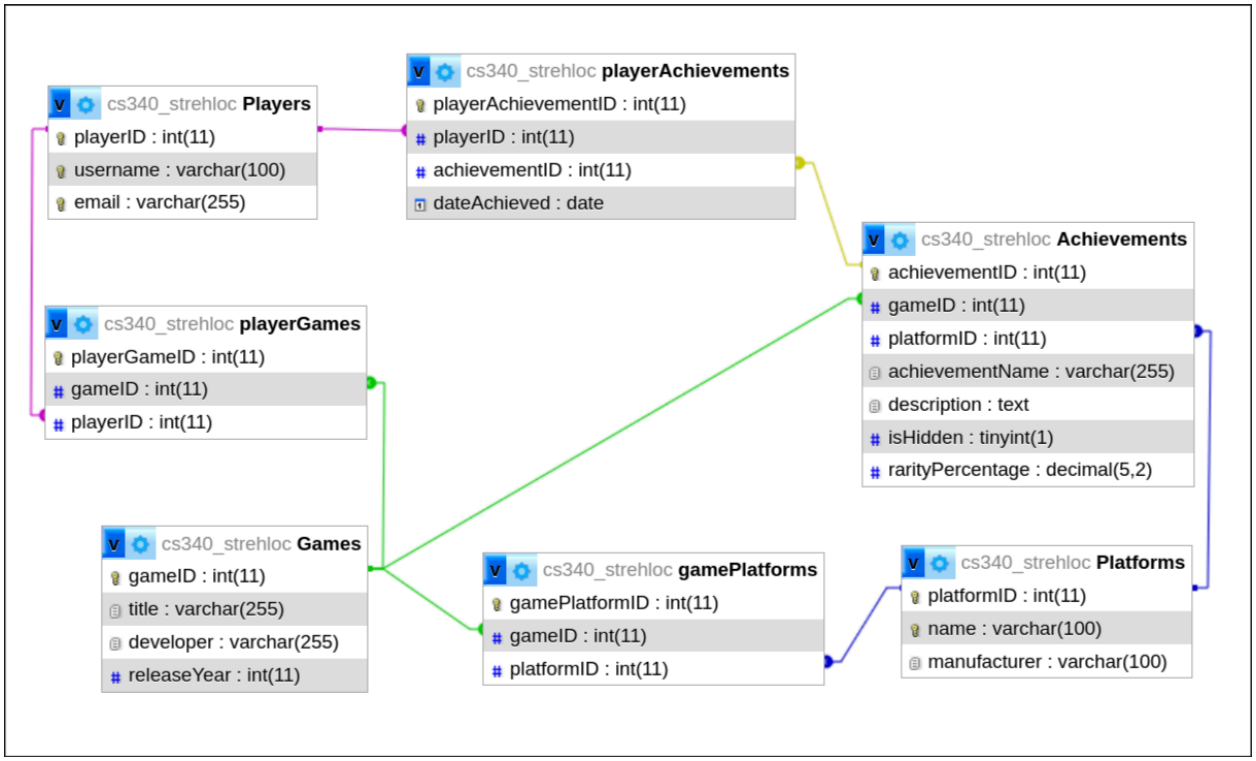
Relationship(s):

- 1:M relationship with Games
- 1:M relationship with Platforms

Entity-Relationship Diagram (ERD)



Schema Diagram



Example Data

Games Table

gameID	title	developer	releaseYear
1	The Legend of Zelda: Breath of the Wild	Nintendo	2017
2	Elden Ring	FromSoftware	2022
3	Stardew Valley	ConcernedApe	2016
4	Hades	Supergiant Games	2020
5	Celeste	Maddy Makes Games	2018

Players Table

playerID	username	email

1	ProGamer99	progamer99@gmail.com
2	TrophyHunter	trophyhunter@gmail.com
3	CasualPlayer	casualplayer@gmail.com
4	SpeedRunner2024	speedrunner@gmail.com

Platforms Table

platformID	name	manufacturer
1	PlayStation 5	Sony
2	Xbox Series X	Microsoft
3	Nintendo Switch	Nintendo
4	Steam	Valve
5	PlayStation 4	Sony

Achievements Table

achievementID	gameID	platformID	achievementName	description	isHidden	rarityPercentage
1	2	1	Elden Lord	Achieve the “Elden Lord” ending	FALSE	45.30
2	2	1	Legendary Armaments	Acquire all 9 legendary armaments	FALSE	5.20
3	2	1	Shardbearer Godrick	Defeat Shardbearer Godrick	FALSE	78.40
4	2	4	Elden Lord	Achieve the “Elden Lord” ending	FALSE	42.10

5	2	4	Legendary Armaments	Acquire all 9 legendary armaments	FALSE	4.80

gamePlatforms Table

gamePlatformID	gameID	platformID
1	1	3
2	2	1
3	2	2
4	2	4
5	3	3

playerGames Table

playerGameID	gameID	playerID
1	2	1
2	2	2
3	3	2
4	3	3
5	4	1

playerAchievements Table

playerAchievementID	playerID	achievementID	dateAchieved
1	1	1	2024-03-15
2	1	2	2024-04-22

3	1	3	2024-03-10
4	2	1	2024-06-12
5	2	3	2024-06-05

Fixes Based on Feedback

Step 1

Link to full reviews: https://edstem.org/us/courses/89768/discussion/7513316

Based on the feedback from classmates, our project would benefit from adding an explicit sentence that describes the issue the database will solve. Some feedback identified a possible issue with including rarityPercentage as an attribute and proposed sourcing this data directly from the platforms to prevent skewed results or removing this attribute as a whole. Additionally, the M:M relationships between games and players, achievements and players, and games and platforms should be implemented with intersection tables to avoid data redundancy. Lastly, there were some name inconsistencies between the ERD and the database outline.

To address the feedback from TAs and classmates, several changes were made to the outline. A sentence that explicitly states the problem the database is meant to solve was added to the top of the overview. Intersection tables (playerGames, playerAchievements, and gamePlatforms) were added to the database outline to manage the multiple M:M relationships, and our ERD was updated to include these. Relationship descriptions were also updated throughout the outline to mention these intersection tables. Additionally, the attribute platformName was changed to name to ensure consistency with the ERD. Lastly, we decided to keep rarityPercentage, with the intention to source the data from the platforms directly.

Step 2

Link to full reviews: https://edstem.org/us/courses/89768/discussion/7625927

Reviews from step 2 mentioned that our sample data should be inserted into the tables using subqueries, so our DDL.sql has been updated for step 3 to do that. It was also suggested that we include more comments for what tables are being inserted into in the DDL, so we updated that as well.

Step 3

Link to full reviews: https://edstem.org/us/courses/89768/discussion/7665579

Reviews from step 3 noted that there was no UPDATE or DELETE for modifying M:M relationships for any of our intersection tables (playerGames, playerAchievements, and gamePlatforms). It was also suggested that we update our CRUD functionality for at least one other entity, as it was mostly limited to Players. Some UI issues were also pointed out, which included a duplicate ‘Hidden Achievement’ field in the Achievement create/update forms (should be a rarityPercentage field), obscure column names for some of the tables, data for username and title columns being swapped in the Players Games table, and the “Update a Game” fields not aligning with the actual Game attributes. Lastly, it was suggested that we avoid using SELECT * to improve UI control.

In response to the feedback, full CRUD functionality was added to every remaining entity table (Games, Platforms, Achievements). Create, Update, and Delete operations were also implemented for all three M:M intersection tables (playerGames, playerAchievements, and gamePlatforms) to allow update and delete operations on the M:M relationships. The create and update forms for Achievements were updated to replace the duplicate ‘Hidden Achievement’ field with ‘rarityPercentage’, obscure column names on all pages were updated to improve readability (e.g., ‘Developer’ to ‘Game Developer’ for Games, ‘Achievement Rarity’ to ‘Rarity Percentage’ in Achievements), corrected the swapped data for ‘username’ and ‘title’ in Player Games issue, and updated the fields for “Update a Game” to align with the actual Game attributes. We also changed ‘SELECT *' to just 'SELECT' as suggested.

Citations

Only original work from Cody Strehlow and Alexa Jauregui is included.